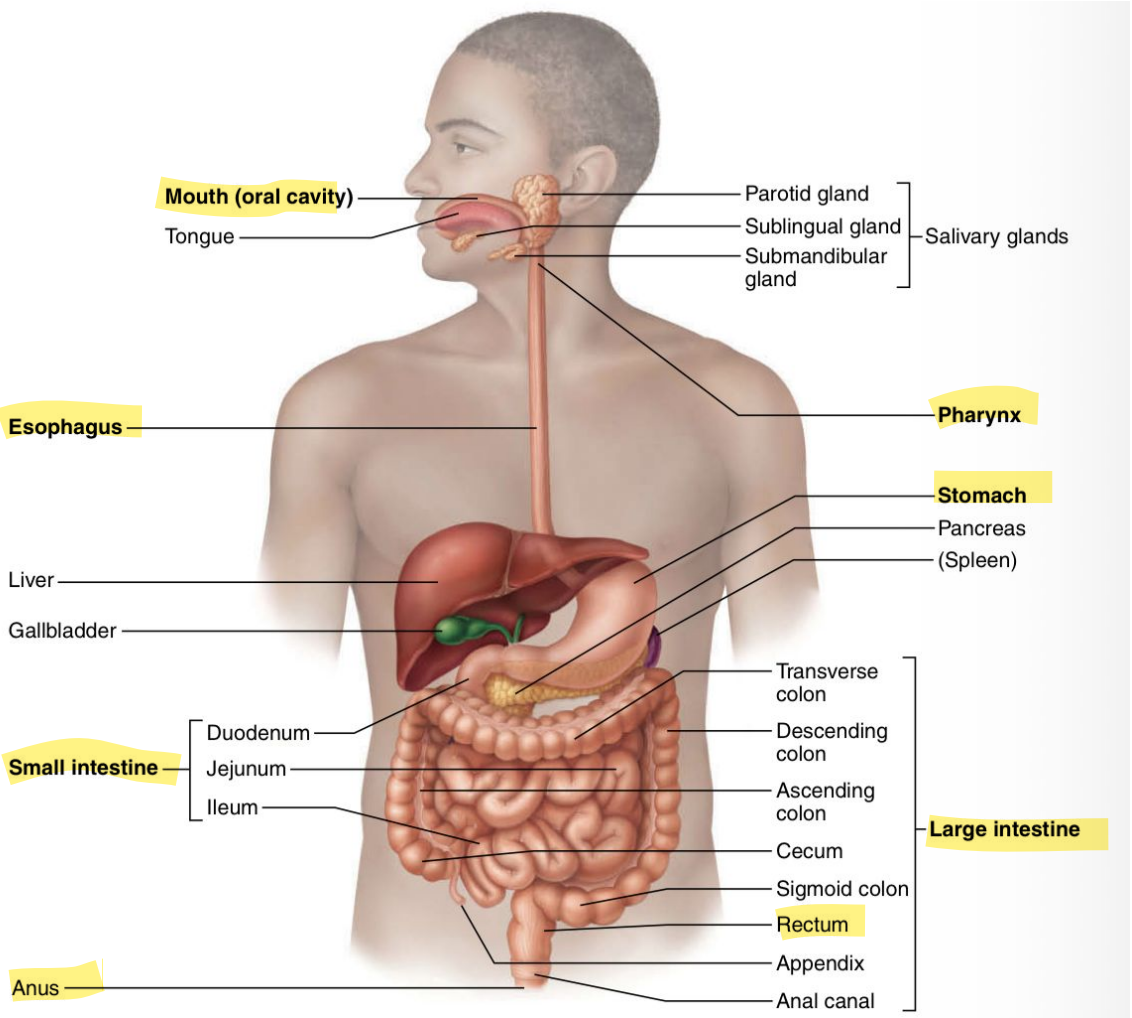
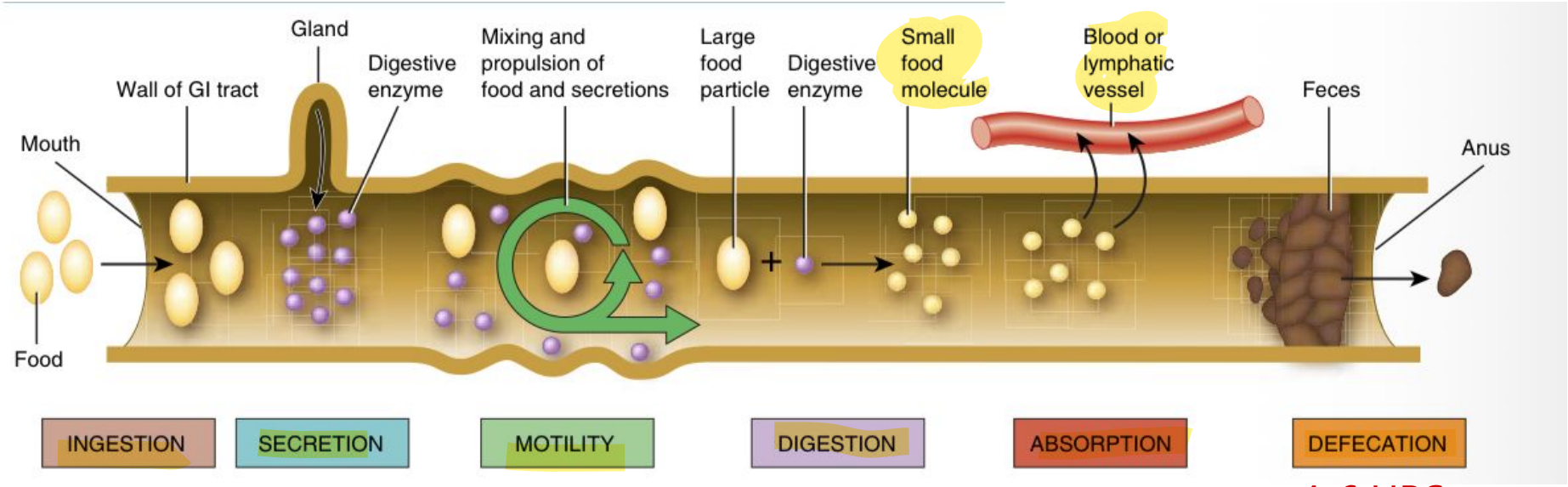
GASTROINTESTINAL DRUGS

Isaac Edron J. Orig, RMT, MD
APMed-1

GI Tract


Mouth (oral cavity)
Tongue
Parotid gland
Sublingual gland
Submandibular gland
Salivary glands
Esophagus
Pharynx
Stomach
Pancreas
(Spleen)
Liver
Gallbladder
Transverse colon
Duodenum
Descending colon
Small intestine
Jejunum
Ascending colon
Ileum
Large intestine
Cecum
Sigmoid colon
Rectum
Appendix
Anus
Anal canal

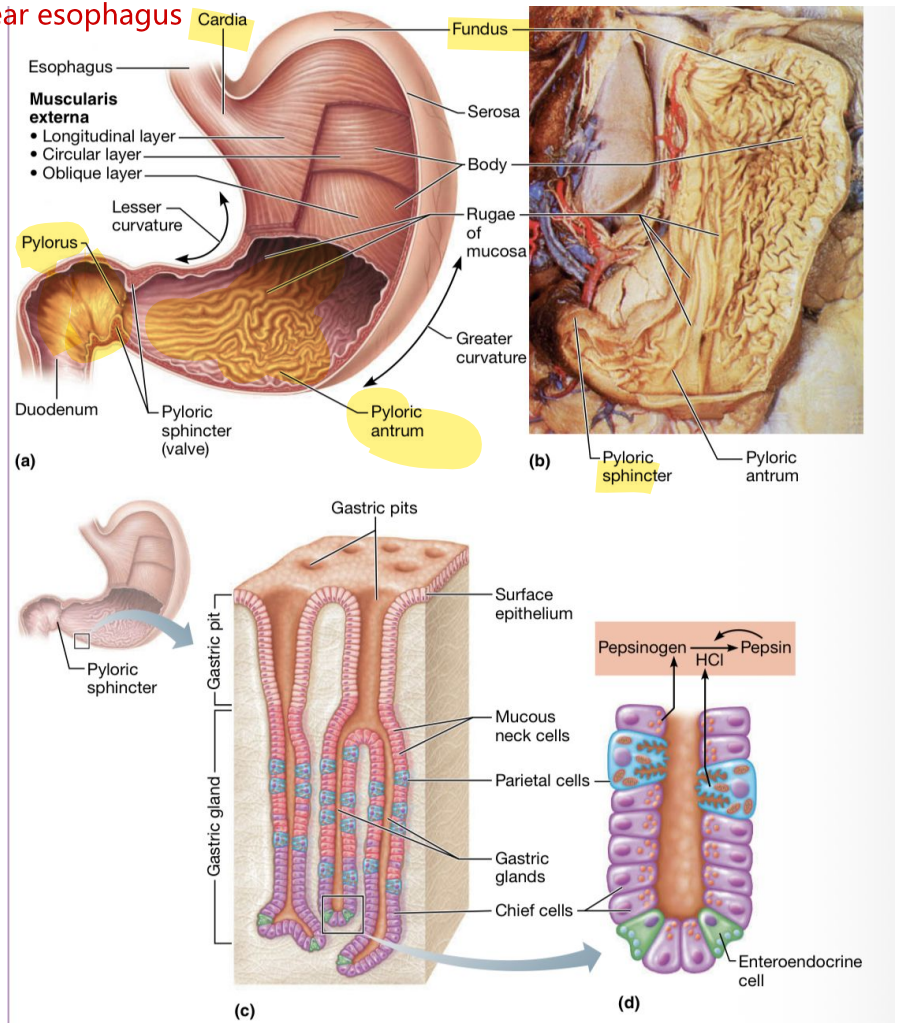


4-6 HRS

ISMDAD

Stomach

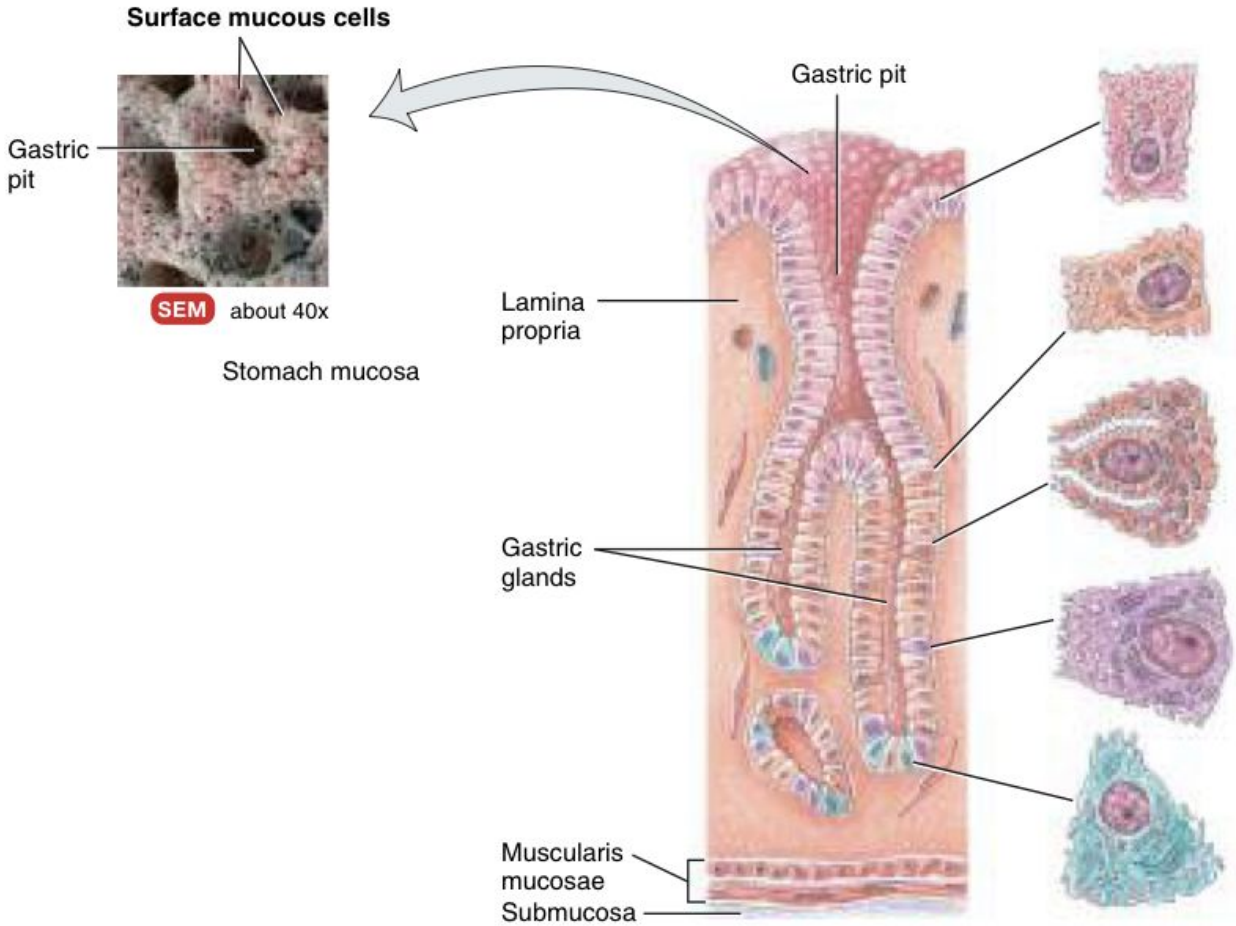
near esophagus



HCLO acid in stomach

sphincter prevents going back to esophagus and directly to duodenum

gastroesophageal juncture prevents gastric acid to go up to esophagus



(b) Sectional view of the stomach mucosa showing gastric glands and cell types

mucus forms lining to protect from acid

used for vitamin B12 absorption and digestion

pepsinogen to pepsin through acid of stomach (active

inactive enzyme that digest proteins

induces acid secretion

Acid Secretion Physiology

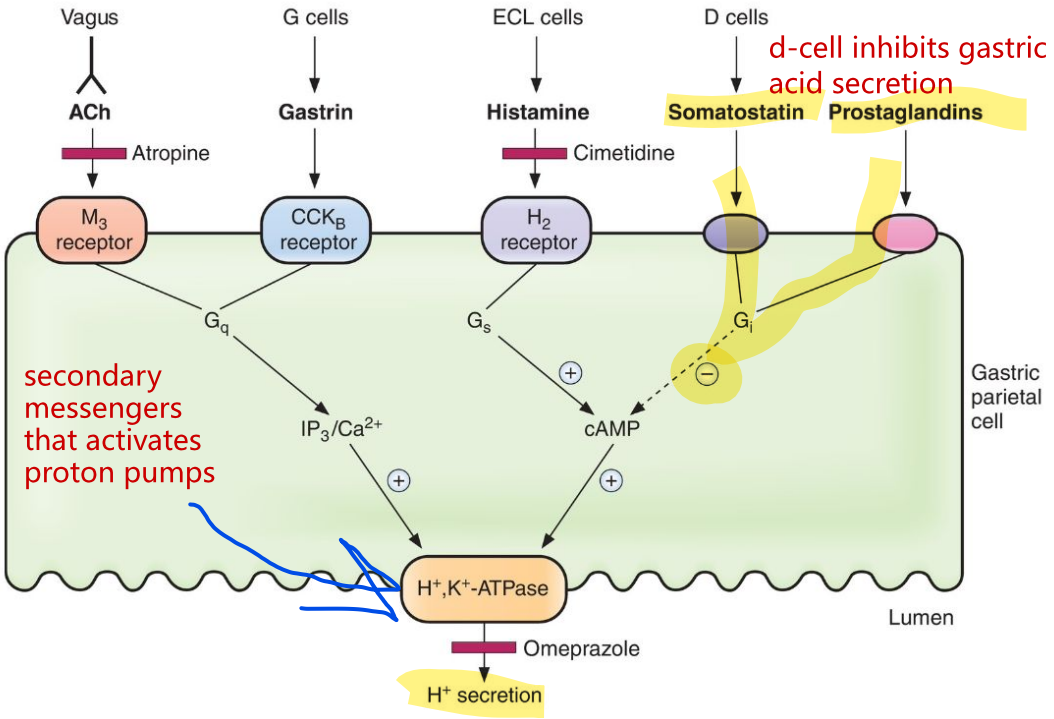
Parietal cell receptors:

ligands

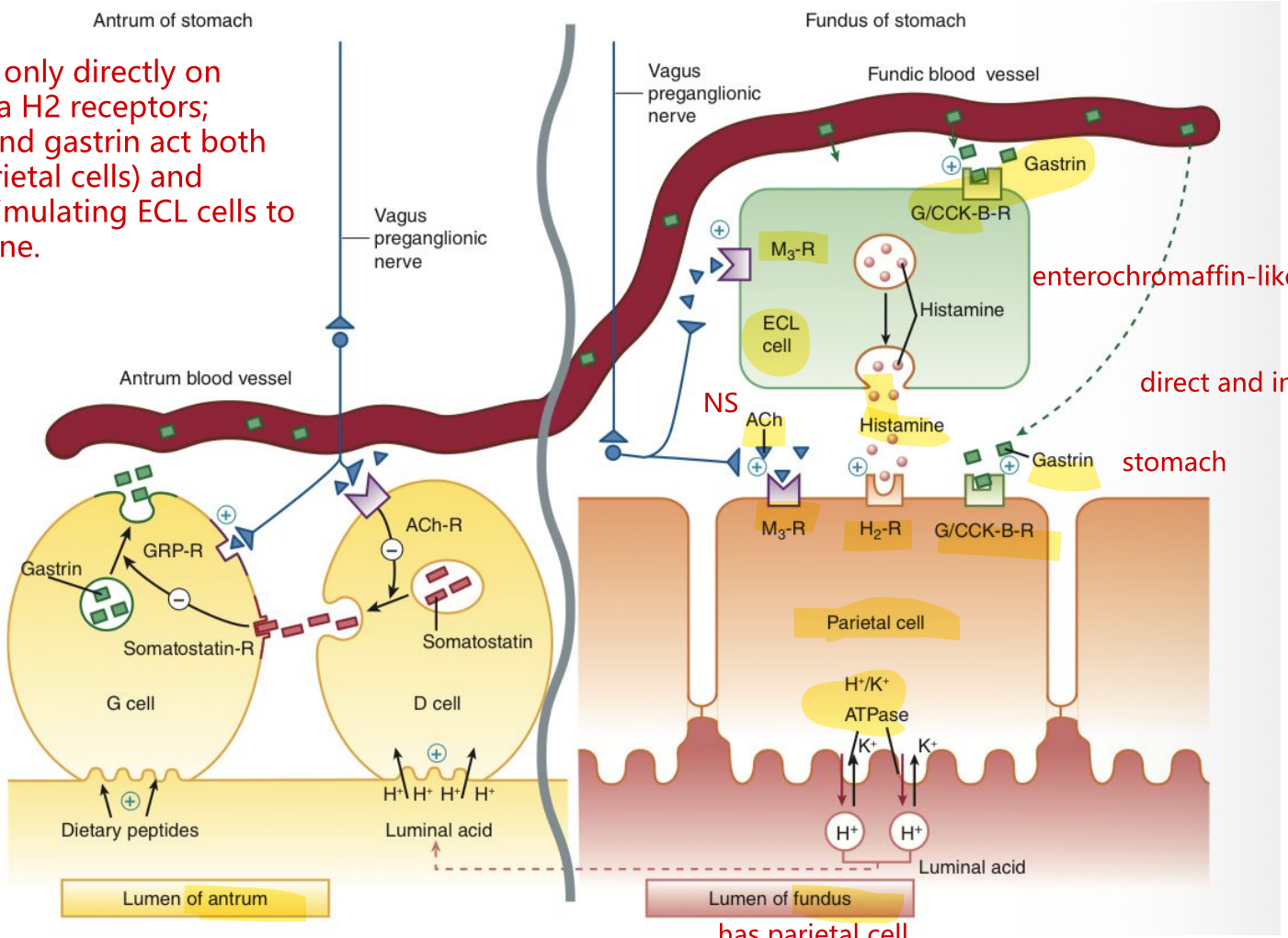
- Gastrin (CCK-B)
- Histamine (H2)
- Acetylcholine (M3)

glands secrete gastric acid (cholinergic - rest and digest)

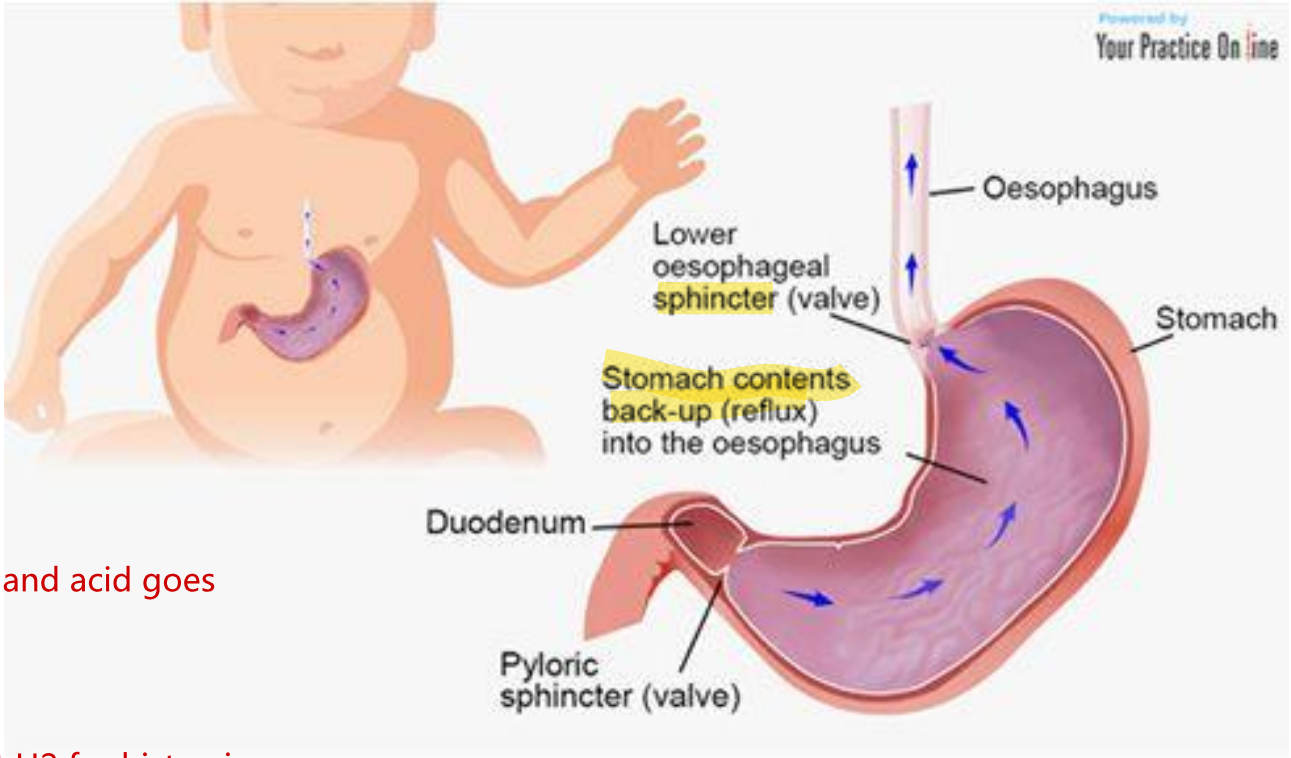
Stimulates acid secretion from the **proton pump (H/K-ATPase)**



Histamine acts only directly on parietal cells via H2 receptors; acetylcholine and gastrin act both directly (on parietal cells) and indirectly by stimulating ECL cells to release histamine.



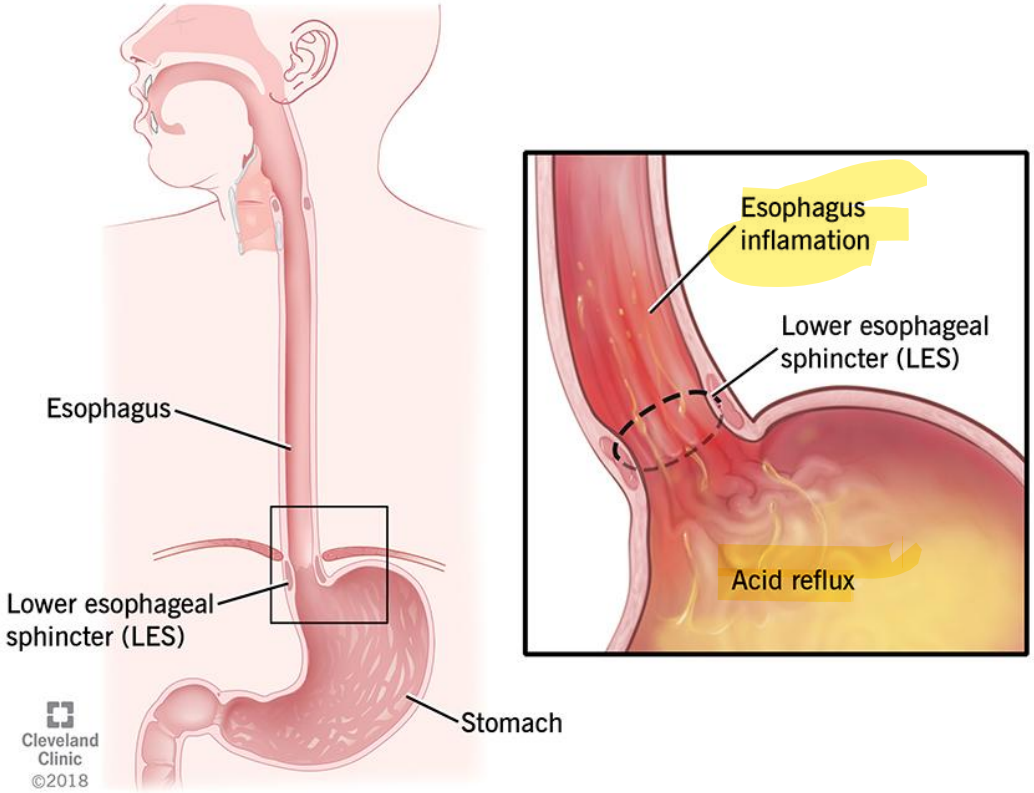
Gastroesophageal Reflux Disease (GERD)



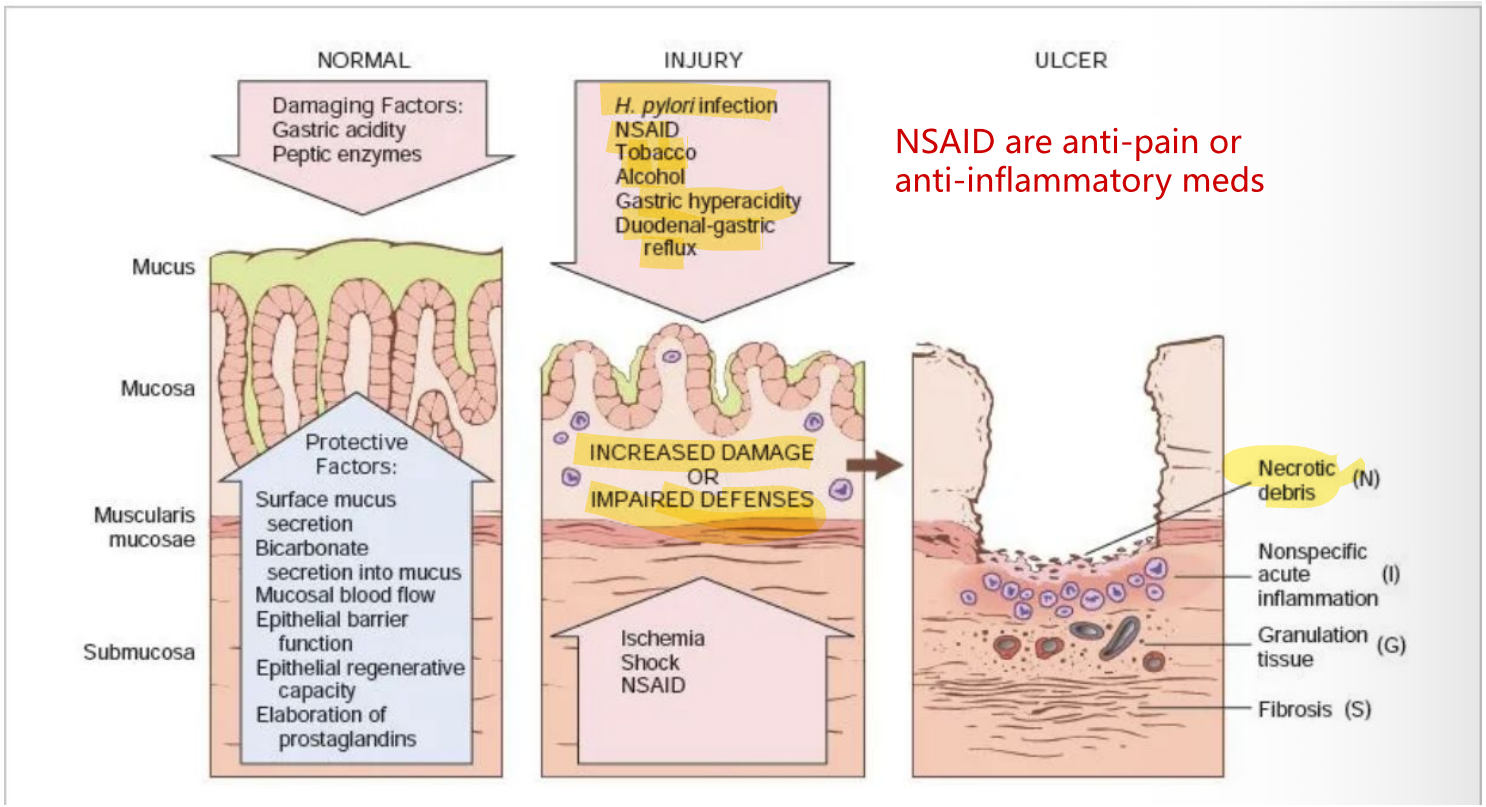
sphincter opens and acid goes to esophagus

allergy is on H1 not H2 for histamine

Gastroesophageal Reflux Disease (GERD)

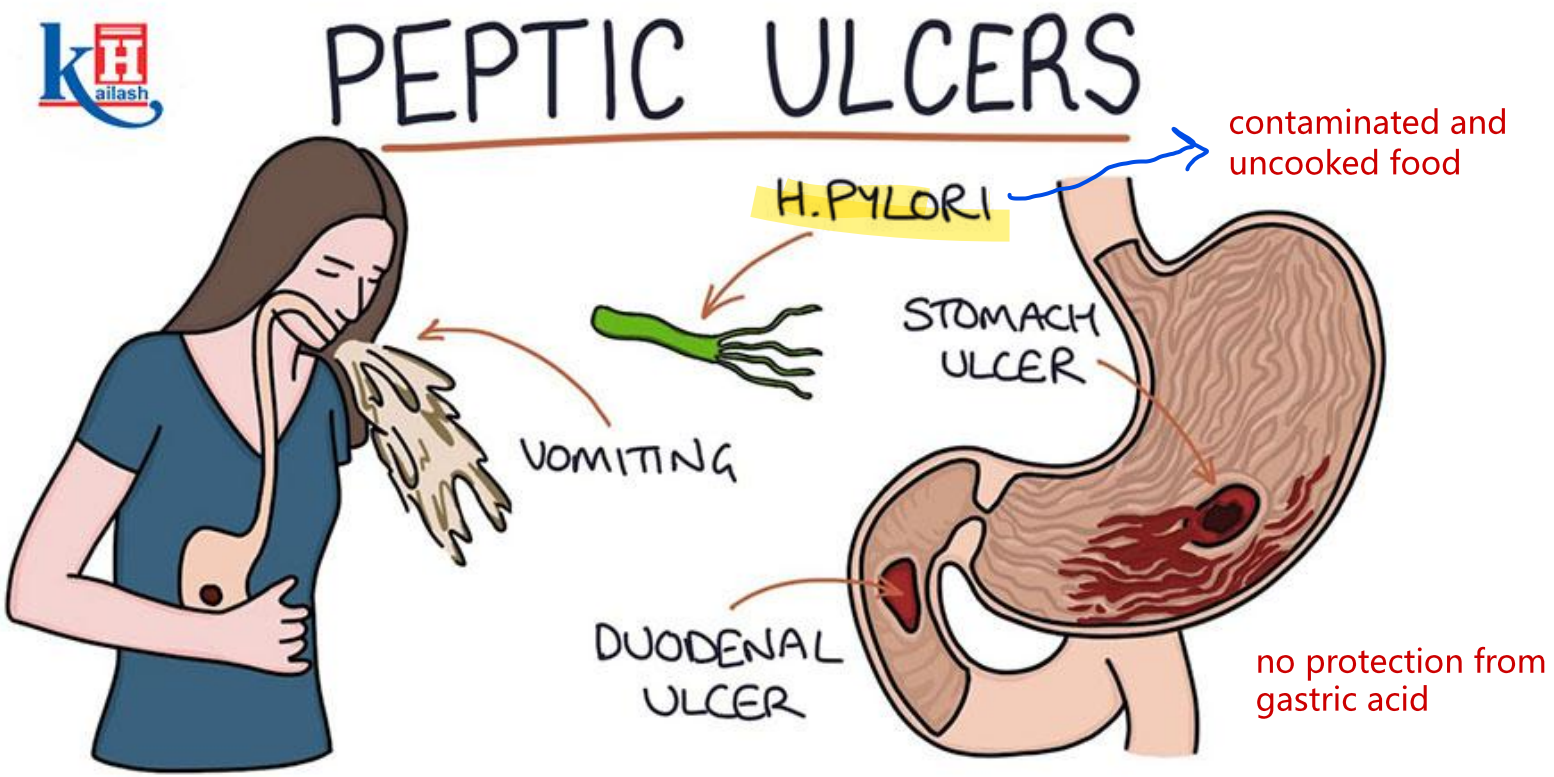


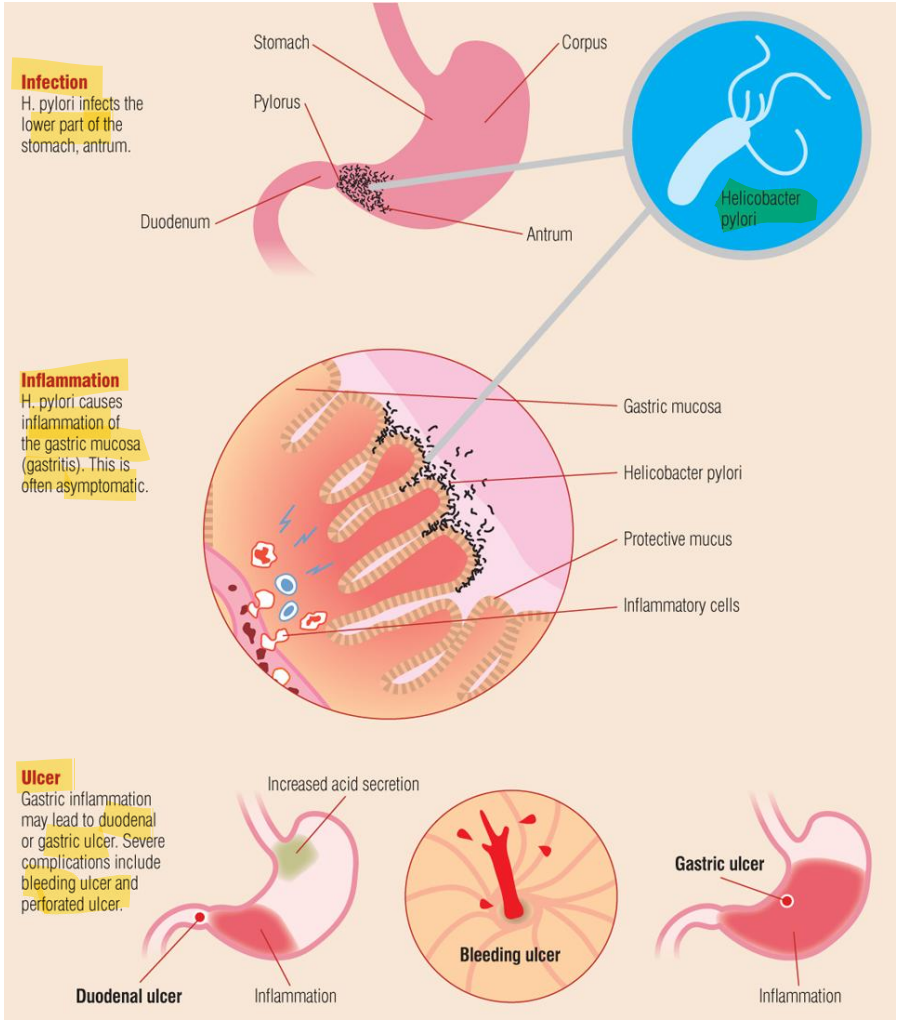
Peptic Ulcer Disease



Peptic Ulcer Disease

ulcer bleeds, puke and poo blood




Infection
H. pylori infects the lower part of the stomach, antrum.
Stomach
Corpus
Pylorus
Duodenum
Antrum
Helicobacter pylori
Inflammation
H. pylori causes inflammation of the gastric mucosa (gastritis). This is often asymptomatic.
Gastric mucosa
Helicobacter pylori
Protective mucus
Inflammatory cells
Ulcer
Gastric inflammation may lead to duodenal or gastric ulcer. Severe complications include bleeding ulcer and perforated ulcer.
Increased acid secretion
Duodenal ulcer
Inflammation
Bleeding ulcer
Gastric ulcer
Inflammation

Drugs for Acid-Peptic Diseases

Drug Classes for Acid-Peptic Diseases

1. Antacids
2. H2-receptor Antagonists
3. Proton Pump Inhibitors (PPI)
4. Sucralfate
5. Prostaglandin Analogs
6. Bismuth Compounds

Antacids

anti-acids

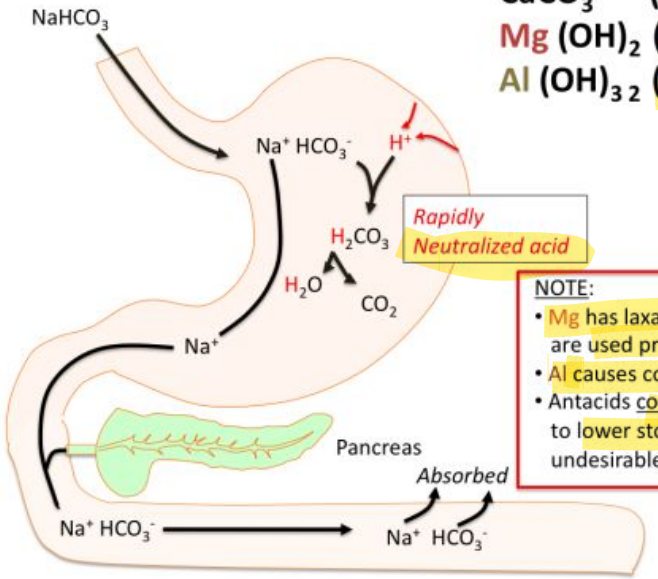
chemical antagonize

Non-prescription remedies for intermittent heartburn and dyspepsia

1. Sodium bicarbonate
2. Calcium carbonate
3. Magnesium hydroxide/Aluminum hydroxide

A single dose of **156 mEq** antacid given **1 hr** after a meal neutralizes gastric acid for up to **2 hours**

Na Bicarbonate



Other Antacids:

$CaCO_3$ (H^+ binds w/ CO_3^{2-})

$Mg(OH)_2$ (H^+ binds w/ OH^-)

$Al(OH)_{3\,2}$ (H^+ binds w/ OH^-)

NOTE:
- Mg has laxative effects (laxatives w/ Mg are used prior to endoscopy)
- Al causes constipation
- Antacids combining Al & Mg are used to lower stomach acid w/o producing undesirable constipation or diarrhea

Adapted from Lullmann H et al. (2005)

Sodium Bicarbonate

Aka **Baking Soda**

Formation of **CO2** results in gastric distention and belching

bloat and burp

Can cause **metabolic alkalosis**

raise pH level makes blood basic

can be give for patients with acidotic



Calcium carbonate

Reacts more slowly than NaHCO3

May also cause belching or metabolic acidosis

dont drink antacids with milk

Na and Ca HCO3 + calcium-containing dairy products = **milk-alkali syndrome**

- hypercalcemia, renal insufficiency, and metabolic acidosis

lower pH level



Magnesium Hydroxide/Aluminum Hydroxide

No gas is generated, no belching

Magnesium = diarrhea

Aluminum = constipation

given together to cancel one another so no GI effects

chemical antagonism



H2 Receptor Antagonists (H2RAs)

Exhibit competitive inhibition at the parietal H2 receptor

histamine would not bind

Suppresses basal and meal-stimulated acid

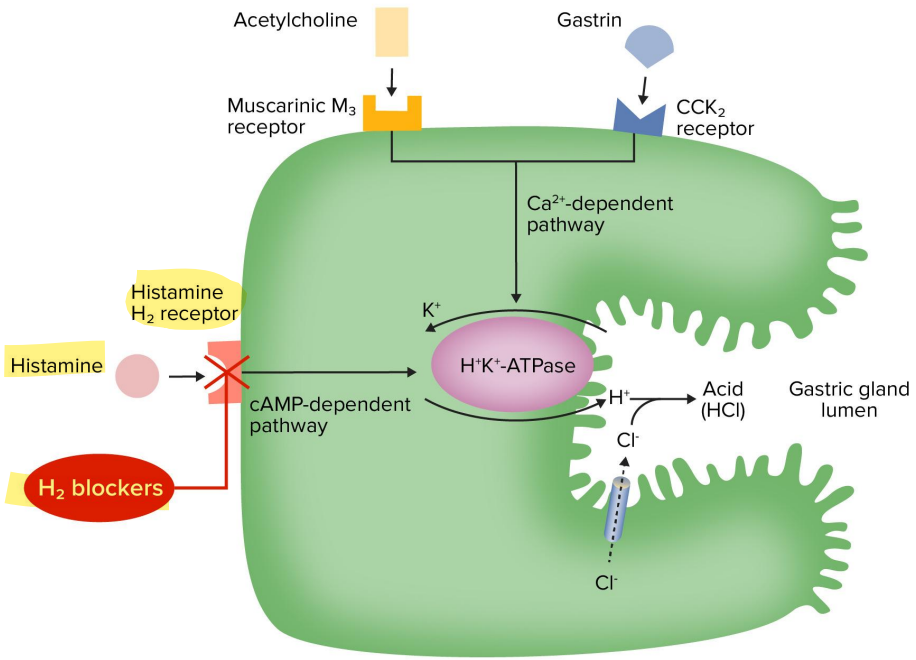
Do not affect H1 or H3 receptors

only H2

1. Cimetidine
2. Ranitidine
3. Famotidine
4. Nizatidine
   - least first-pass metabolism

-TIDINE
CRFN

drug avoids significant breakdown by the liver when taken orally, so more of it reaches the bloodstream in active form



Drug	Relative Potency	Dose to Achieve >50% Acid Inhibition for 10 Hours	Usual Dose for Acute Duodenal or Gastric Ulcer	Usual Dose for Gastroesophageal Reflux Disease	Usual Dose for Prevention of Stress-Related Bleeding
Cimetidine	1	400–800 mg	800 mg HS or 400 mg bid	800 mg bid	50 mg/h continuous infusion
Ranitidine	4–10	150 mg	300 mg HS or 150 mg bid	150 mg bid	6.25 mg/h continuous infusion or 50 mg IV every 6–8 h
Nizatidine	4–10	150 mg	300 mg HS or 150 mg bid	150 mg bid	Not available
Famotidine	20–50	20 mg	40 mg HS or 20 mg bid	20 mg bid	20 mg IV every 12 h

bid, twice daily; HS, bedtime.

famotidine most potent only need small amount

cimetidine is least potent

H2 Receptor Antagonists (H2RAs)

Clinical Uses:

Dyspepsia, also known as indigestion, is a condition marked by discomfort or pain in the upper abdomen, often after eating or drinking.

1. **GERD** h2ras and antacids
   - infrequent heartburn or dyspepsia (<3x per week)
2. **Peptic Ulcer Disease**
   - largely replaced by PPIs
3. **Nonulcer dyspepsia**
4. **Prevention of bleeding from stress-related gastritis**



inflammation of the stomach lining, which can cause pain, nausea, bloating, and indigestion.

H2 Receptor Antagonists (H2RAs)

Extremely safe drugs

Adverse Effects

1. Mental status changes (cimetidine)
2. Gynecomastia, impotence in men (cimetidine)



gynecomastia is man boobs
cimetidine increases estrogen levels on men

Proton-Pump Inhibitors

targets proton pumps

Administered as inactive **prodrugs**

Administered approximately **1 hour** before a meal (usually breakfast)

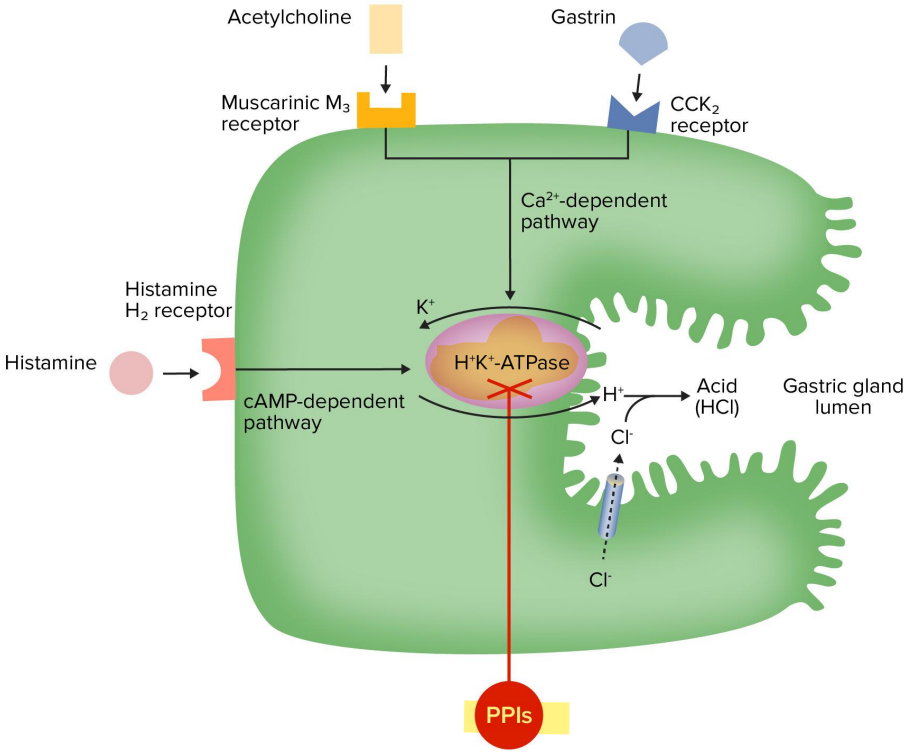
once a day

Acid inhibition lasts up to **24 hours** owing to **irreversible inactivation** of the proton pump

1. Omeprazole
2. Esomeprazole
3. Lansoprazole
4. Dexlansoprazole
5. Rabeprazole
6. Pantoprazole

-PRAZOLE

DROPLE



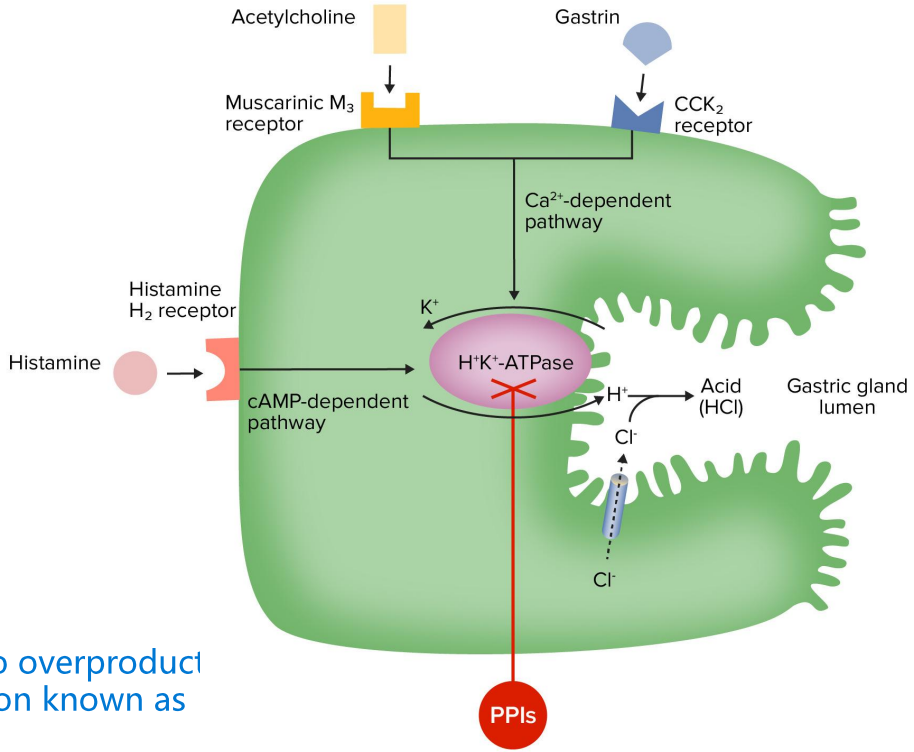
body needs to make new hk atpase
ppis are now mostly given

Proton-Pump Inhibitors

Clinical uses:

1. **GERD**
   - Most effective agents
   - Usually intermittent courses
2. **PUD**
   - more rapid symptom relief and faster ulcer healing than H2RAs
3. **Nonulcer dyspepsia**
4. **Prevention of stress-related mucosal bleeding**
5. **Gastrinoma**

rare tumor that secretes excessive gastrin, leading to overproduct of stomach acid and severe peptic ulcers—a condition known as Zollinger-Ellison syndrome



common reason of ulcer

Helicobacter pylori Regimens

ppi is used for healing

H pylori Treatment Regimens

Therapy	Dosing	Duration (days)[a]	Eradication (%)
	First-Line Options		
Clarithromycin triple	PPI[b] bid Clarithromycin 500 mg bid Amoxicillin 1 g bid *or* Metronidazole 500 mg tid	14	70-85
Bismuth quadruple[c]	PPI bid Bismuth subcitrate 120-300 mg/ bismuth subsalicylate 300 mg qid Metronidazole 250 or 500 mg qid Tetracycline 500 mg qid	10-14	75-90
Nonbismuth-based quadruple	PPI bid Amoxicillin 1 g bid Clarithromycin 500 mg bid Metronidazole 500 mg bid	10-14	90

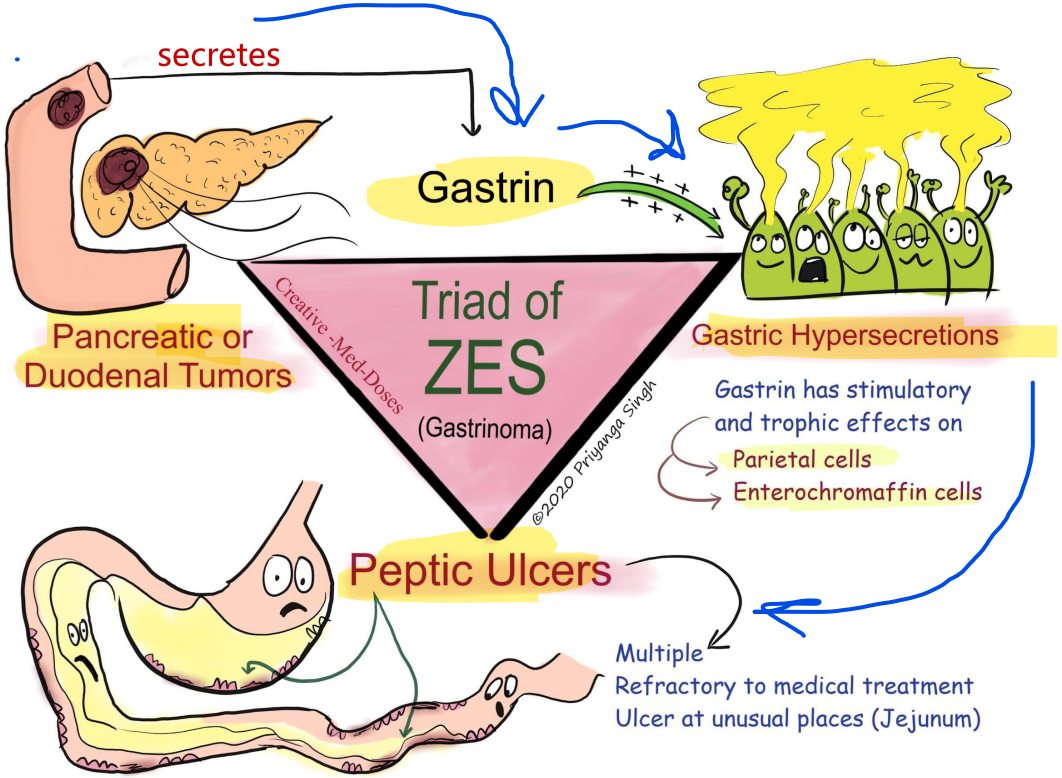
antibacterial agents

what we give since available on PH

Gastrinoma

zollinger ellison syndrome

surgery is done to remove tumors and give PPI



Proton-Pump Inhibitors

Adverse Effects:

1. Vitamin B12 deficiency
2. Increased respiratory and enteric infections

gastrin acid is a protective layer by body against microbes



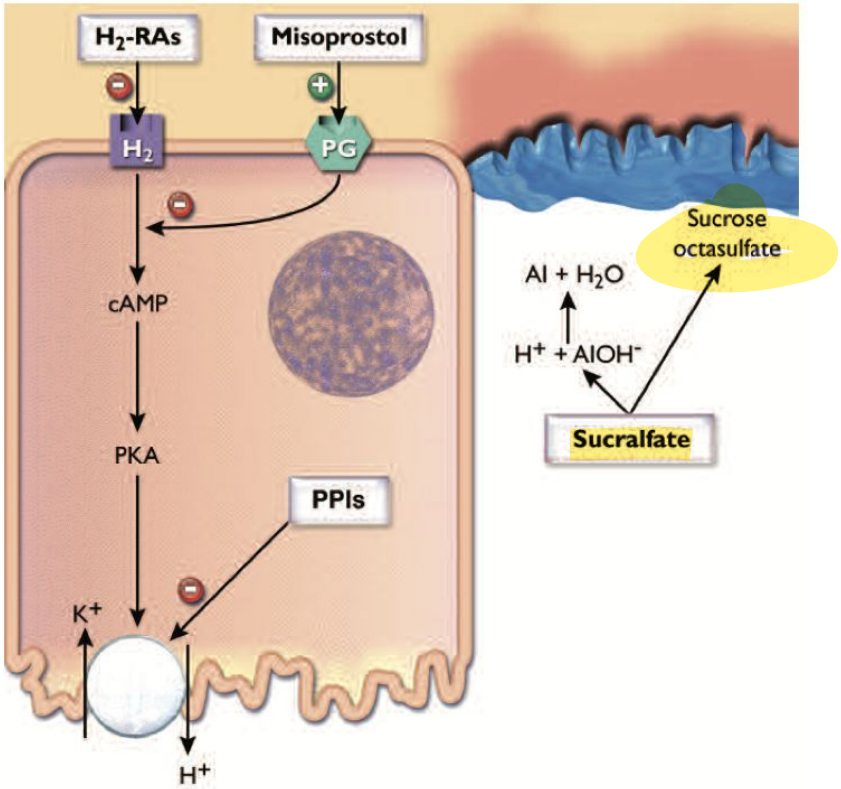


Sucralfate

Salt of sucrose complexed to sulfated aluminum hydroxide

Forms **viscous, tenacious paste** that binds selectively to ulcers or erosions for up to **6 hours**

Administered 1g 4x daily on an empty stomach (1 hr before meals)



Prostaglandin Analogs

inhibitors of gastric acid

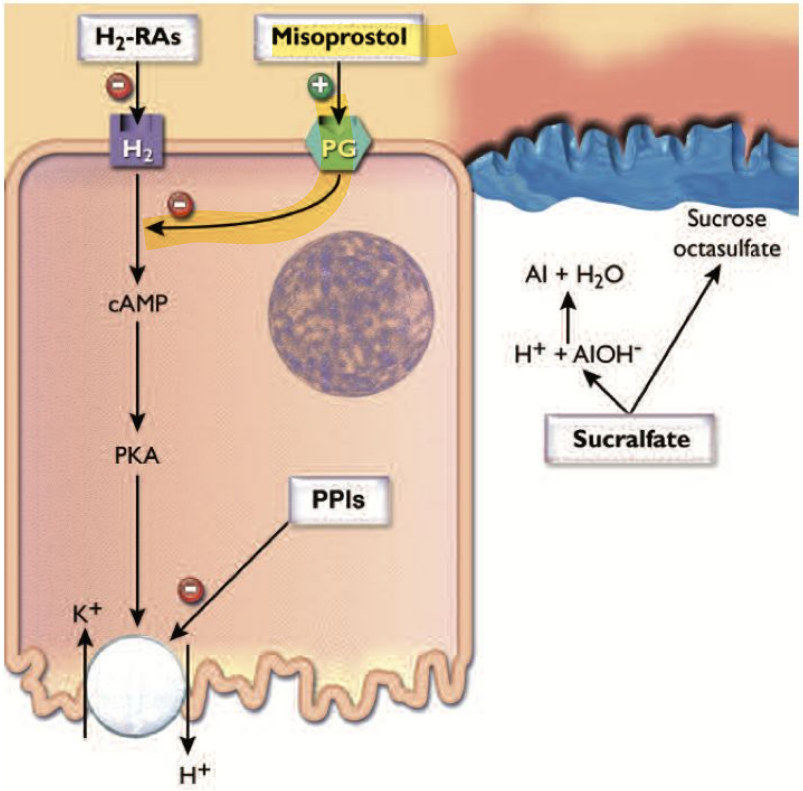
Misoprostol (analog of PGE2)

decreases gastric acid secretion
can cause abortion dont give for pregnant

Administered 3-4x daily

Can reduce incidence of NSAID-induced ulcers

NSAID stands for Non-Steroidal Anti-Inflammatc
Drug—a class of medications used to reduce pai
inflammation, and fever.

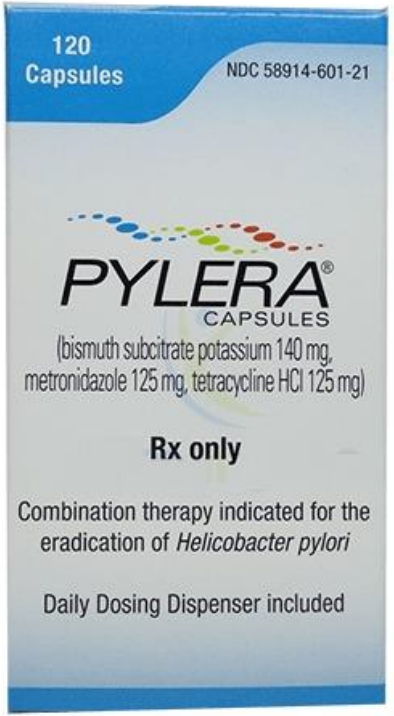


Bismuth Compounds

Coats ulcers and erosions

Stimulate prostaglandin, mucus, and bicarbonate secretion pmb

Has direct antimicrobial activity against H. pylori



Laxatives

Laxatives

given for px with constipation to poop

1. Bulk-forming laxatives
2. Stool softeners
3. Osmotic laxatives
4. Stimulant laxatives

Bulk-Forming Laxatives

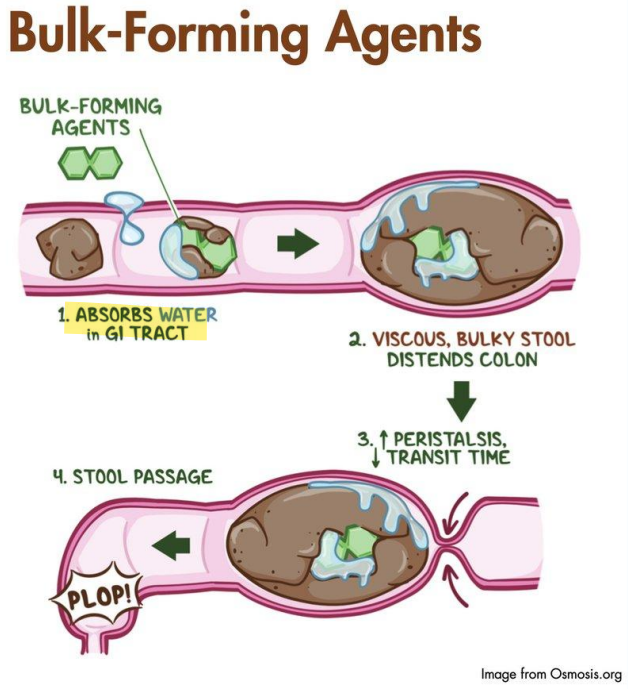
Absorb water

Forming a **bulky, emollient gel** that **distends the colon and promotes peristalsis**

stimulated to contract and poop

PiMP

1. Psyllium
2. Methylcellulose
3. Polycarbophil



Active Ingredient
Psyllium husk; indigestible, fibrous compounds derived from plants

How it Works?
- These agents are not absorbed by the intestines and result to 'bulking up' of the stool
- Softens and increases the bulk of digested food, resulting to colonic distention and increased peristalsis until stools are passed out
- Generally produces effect in 12-72 hours

Considerations
- Can ferment in the intestine, which may cause gas and bloating
- Take with fluids for bulk-forming agents to work and to avoid fecal impaction

about 5 days of constipation use laxatives

Stool Softeners

Soften stool material, permitting water and lipids to penetrate

May be administered orally or rectally

DGM

1. Docusate (oral or enema)
2. Glycerin suppository
3. Mineral oil

stimulant free softener





Osmotic Laxatives

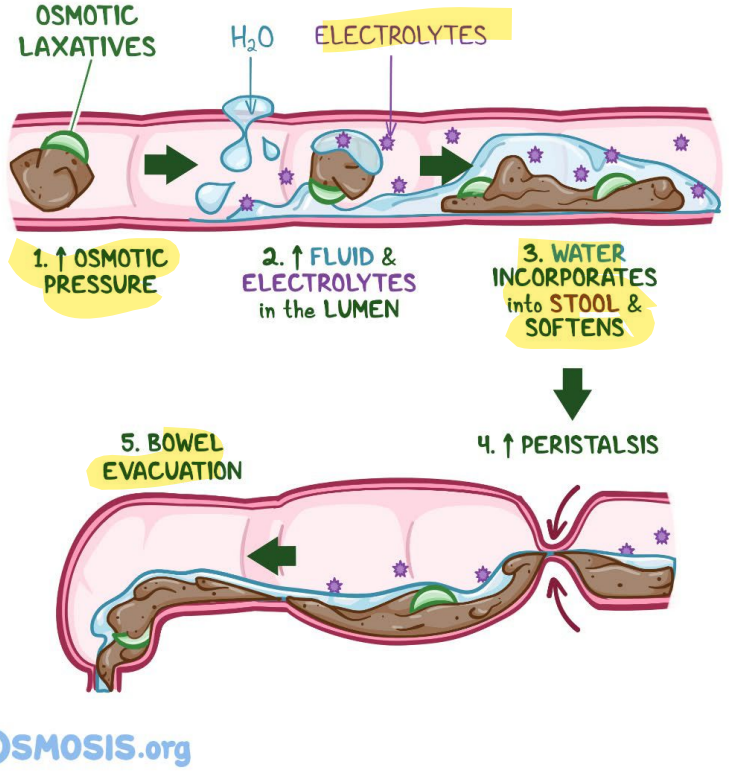
Soluble but nonabsorbable compounds that result in **increased stool liquidity due to an obligate increase in fecal fluid**

more water causes large intestines to release poop

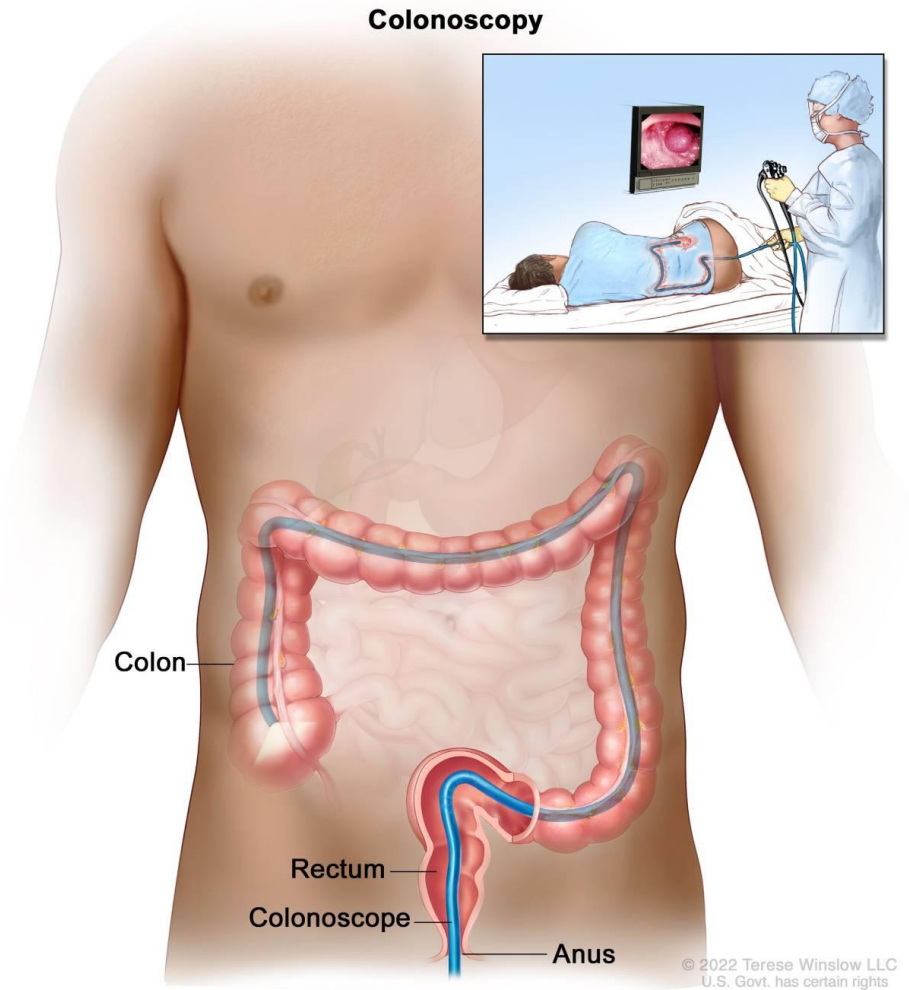
LP

1. Lactulose
2. Polyethylene Glycol (PEG)
   - used for complete colonic cleansing before GI endoscopy

give PEG and dulcolax




7 Sachets x 17 g
(Net Wt. 119 g)
Polyethylene Glycol 3350
SureLax®
17 g Powder for Oral Solution
Laxative
Net Wt. 17 g
Polyethylene Glycol 3350
SureLax®
17 g Powder for Oral Solution
Laxative


Colonoscopy
Colon
Rectum
Colonoscope
Anus
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Stimulant Laxatives

1. **Anthraquinone derivatives**
   - Aloe, senna, cascara
   - cause "**melanosis coli**"
2. **Diphenylmethane derivatives**
   - bisacodyl
   - induces bowel movement
     - 6-10hrs (orally)
     - 30-60mins (rectally) insert anus

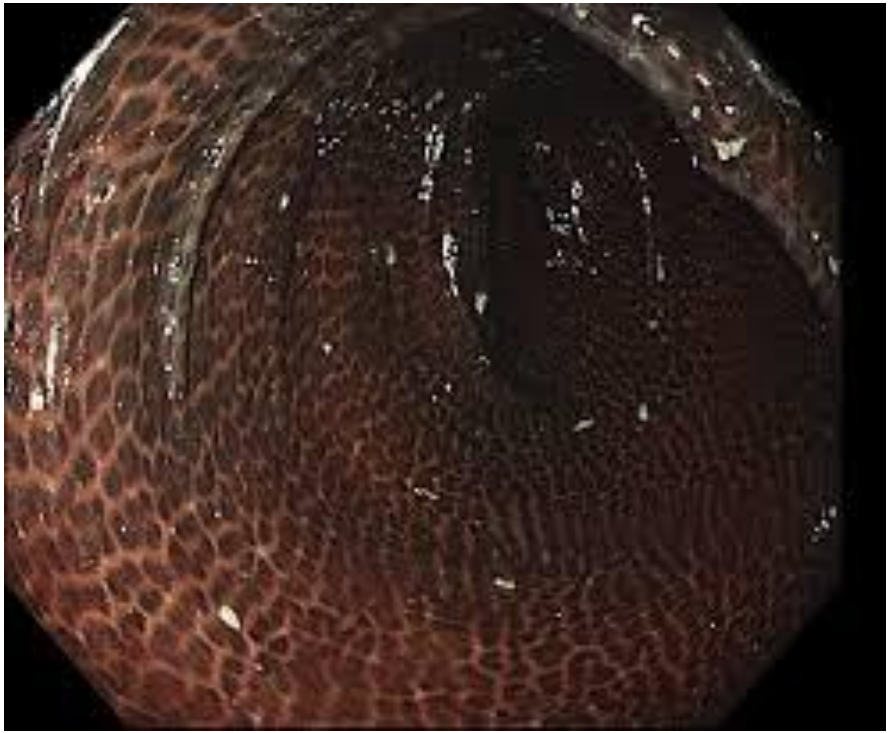
herbal laxatives used to relieve constipation, but they can cause side effects like cramping, dehydration, and electrolyte imbalance if overused



stimulant laxative senna



melanosis coli - black melanin colon



chronic senna taker has colon will have hyperpigmentation

Thank you!